

TOUSSUS-LE-NOBLE AP, France

WMO: 071460

Lat: 48.7506N

Lon: 2.1133E

Elev: 538

StdP: 14.41

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.3	26.2	16.0	12.6	28.2	20.1	15.3	28.8	33.4	49.6	27.7	47.4	5.6	30	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	87.8	68.1	83.9	66.8	80.3	65.3	70.0	83.7	68.2	80.0	66.6	77.5	8.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	96.5	74.7	63.9	90.7	72.8	62.4	85.9	71.1	34.3	83.6	32.7	80.1	31.4	77.6	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	22.1	18.7	18.2	95.8	5.0	4.6	14.6	99.1	11.7	101.7	8.9	104.3	5.3	107.6
			17.4	73.6	5.0	1.9	13.8	75.0	10.9	76.1	8.1	77.1	4.5	78.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	39.2	40.2	45.5	51.0	57.2	63.3	66.8	66.0	60.5	53.7	45.3	40.2	(6)
(7)		DBStd	11.66	6.90	6.75	6.25	6.20	6.01	6.03	5.93	5.59	5.28	6.23	6.16	6.76	(7)
(8)		HDD50	1357	336	277	166	59	9	1	0	0	1	35	165	308	(8)
(9)		HDD65	4837	799	694	605	421	252	102	47	52	154	351	592	768	(9)
(10)		CDD50	2261	2	3	26	89	232	400	521	495	315	150	23	5	(10)
(11)		CDD65	268	0	0	0	1	10	52	103	82	19	1	0	0	(11)
(12)		CDH74	2762	0	0	0	26	123	523	1053	838	188	11	0	0	(12)
(13)		CDH80	848	0	0	0	1	15	154	352	289	36	1	0	0	(13)
(14)	Wind	WSAvg	8.7	10.0	10.6	9.9	9.0	8.5	7.8	7.6	7.0	7.0	8.2	9.1	10.0	(14)
(15)	Precipitation	PrecAvg	26.5	2.3	1.9	2.2	1.9	2.4	2.1	2.4	2.0	2.1	2.4	2.4	2.3	(15)
(16)		PrecMax	33.6	4.7	4.2	4.3	4.4	4.6	6.3	6.0	4.5	5.0	6.5	3.9	4.6	(16)
(17)		PrecMin	16.3	0.5	0.3	0.1	0.1	0.3	0.1	0.6	0.1	0.3	0.3	0.4	0.2	(17)
(18)		PrecStd	4.4	1.1	1.1	1.1	1.1	1.2	1.2	1.3	1.1	1.3	1.7	0.9	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	59.7	68.4	77.4	81.8	90.2	93.5	95.1	84.4	75.3	62.1	56.5	(19)
(20)		MCWB	52.9	50.4	53.6	60.4	66.9	70.8	69.8	70.8	66.0	63.3	57.7	53.4	(20)	
(21)		2%	DB	53.2	54.4	63.6	72.4	77.4	84.3	87.6	87.7	79.6	69.8	59.3	54.3	(21)
(22)		MCWB	50.7	48.9	52.5	57.6	62.6	67.4	67.6	68.0	64.3	61.6	56.0	51.7	(22)	
(23)		5%	DB	51.1	52.0	59.3	66.9	73.6	79.7	83.7	82.6	75.1	65.7	56.6	52.2	(23)
(24)		MCWB	48.9	48.5	50.8	55.1	61.1	65.4	66.8	65.9	63.3	59.2	53.6	49.9	(24)	
(25)		10%	DB	49.0	50.0	55.7	62.8	69.4	75.8	80.0	78.3	71.3	62.7	54.2	50.2	(25)
(26)		MCWB	46.9	46.9	49.6	53.1	58.8	63.8	65.4	64.0	61.0	57.8	51.6	48.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.6	52.3	55.9	62.1	67.9	73.1	71.6	72.1	68.2	64.9	59.2	54.1	(27)
(28)		MCDWB	55.3	55.8	63.5	74.6	79.4	87.5	88.1	89.8	79.1	71.1	60.8	55.5	(28)	
(29)		2%	WB	51.2	50.4	53.9	58.9	64.9	69.2	69.7	69.4	66.0	62.1	56.2	52.1	(29)
(30)		MCDWB	52.7	53.0	60.1	70.1	75.0	80.1	83.6	83.5	76.1	67.7	58.8	53.9	(30)	
(31)		5%	WB	49.1	48.9	52.2	56.2	62.1	66.9	68.1	67.3	64.1	60.3	53.9	50.4	(31)
(32)		MCDWB	50.9	51.4	57.7	65.2	70.5	76.8	80.3	79.2	72.6	65.1	56.2	52.2	(32)	
(33)		10%	WB	47.0	47.1	50.4	53.9	59.9	64.8	66.3	65.5	62.5	58.6	51.9	48.3	(33)
(34)		MCDWB	48.7	49.8	55.2	61.0	67.4	74.2	77.7	75.9	69.4	62.0	54.1	50.1	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.9	11.7	15.2	17.9	18.5	19.4	20.2	20.5	18.9	14.4	10.4	9.2	(35)	
(36)		5% DB	MCDBR	9.5	14.5	21.7	24.7	24.6	26.3	27.0	28.2	25.0	17.7	11.9	9.3	(36)
(37)		MCWBR	8.3	10.5	12.5	12.6	12.2	12.4	11.2	11.9	11.8	10.6	9.1	8.1	(37)	
(38)		5% WB	MCDBR	9.2	12.8	17.7	22.3	21.8	23.1	24.3	24.3	21.2	15.8	11.3	9.2	(38)
(39)		MCWBR	8.4	10.2	11.3	12.2	12.0	12.3	10.9	11.6	11.1	10.2	9.3	8.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.332	0.353	0.387	0.404	0.411	0.413	0.412	0.401	0.375	0.370	0.342	0.326	(40)	
(41)		taud	2.389	2.342	2.241	2.213	2.232	2.263	2.261	2.311	2.372	2.373	2.394	2.361	(41)	
(42)		Ebn at Noon	228	250	259	267	270	269	268	265	261	240	221	212	(42)	
(43)		Edh at Noon	22	28	36	42	43	42	41	37	32	27	21	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	291	539	912	1383	1630	1819	1723	1498	1157	683	352	245	(44)	
(45)		RadStd	33	75	113	156	155	171	140	144	94	55	34	37	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page